Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	
Project Name	Health360: Executive Analytics & Population Health Dashboard
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S. No.	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyse data.	Tableau Calculated Fields (Calculated Syntax)	Used to construct dynamic business logic including the Age Category chronological classification (Children to Elder), Calculated Risk Score, and binary aggregations for chronic diseases.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	Flat File CSV (smart_healthcare_dataset.csv)	Stores 5,000 anonymized patient records containing clinical vitals, lifestyle metrics, and population demographic variables in a structured format.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Tableau Native Data Connector	Establishes a direct file connection to ingest the 5,000-row healthcare CSV into the Tableau engine without data loss or schema distortion.
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Tableau Desktop / Tableau Data Extract (.hyper)	Converts raw CSV records into an optimized columnar .hyper extract, executing pre-calculated measures and dimensions for sub-second visual query execution.

5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau Desktop Professional	Designed 9 core analytical worksheets (scatter plots, heatmaps, stacked bars, pie charts), a single-page Master Dashboard, and a 9-point executive Tableau Story.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public Web Server / Cloud Engine	Delivers cloud-hosted, zero-footprint web access for executive personas (Dr. Carter, Emily Rodriguez, Dr. Priya Nair) with full support for global cross-filtering.